

COSC 1030 Penny Searching Exercise

Goal:

The goal of this activity is to start to learn vocabulary associated with Parallel/Distributed Processing (PDC) and begin to build intuition about how to use these concepts to improve algorithm implementation.

Materials Needed:

A container with 20 coins of all the same denomination.

A Note to Online Students:

Normally this activity is best suited to a group of 3-5 people. Since you are on-line people, you can either grab some friends or family members to be part of your group or you will just have to imagine having multiple people! No prior knowledge is required about computing to participate in this activity and it will definitely be more fun with a group of people.

Phase 1: Sequential

Select one person to be the sorter (Person 1) and another person to be the timer (Person 2).

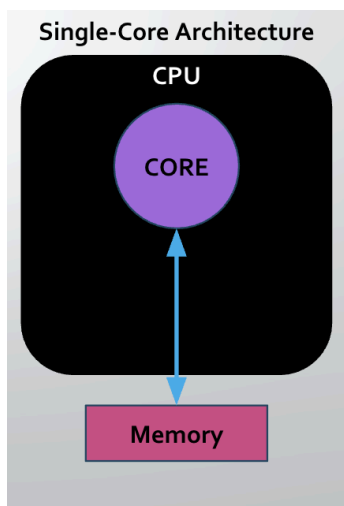
Person 1 performs sorting through a single bag of coins to find the oldest coin.

Person 2 times the work of Person 1 and then writes the time down for finding the oldest coin.

All other people observe.

What did PHASE 1 represent?

This phase represents a purely **sequential** algorithm where there is only one **processor** (or worker) performing the task. There is no parallelism, and the task is completed by a single worker from start to finish. This is akin to a **single-core processing** model in computing where no parallel execution takes place.



Question about this phase:

If we were to divide the coins into two groups in the beginning and have two processors (people) each looking for the oldest coin at the same time, do you think this would take more or less time?

Phase 2: Parallel Version A

Select two people (Person 1 & Person 2) to be the sorters. Both people should be given a bag of half the coins each.

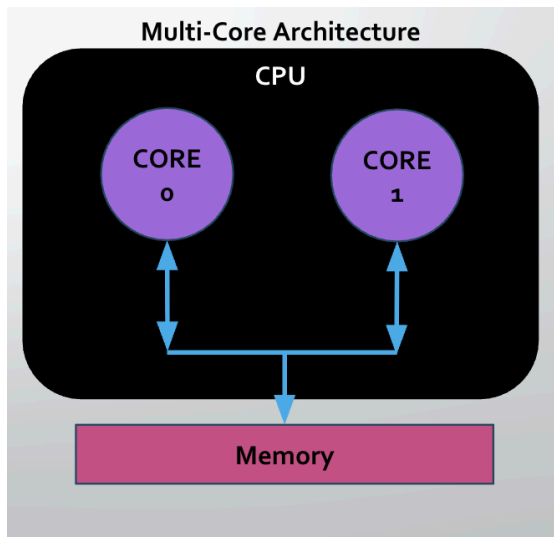
Person 1 performs sorting through their bag of coins to find the oldest coin. Simultaneously (at the same time), Person 2 performs sorting through their single bag of coins to find the oldest coin.

Then, Persons 1 & 2 compare their oldest coin and select the oldest between the two coins.

Person 3 records the processing time of finding the oldest coin on the group lab sheet.

What did PHASE 2 represent?

In this phase, two independent tasks (searching two separate bags) are executed **concurrently**. This represents **task parallelism**, where multiple workers perform tasks in parallel and then synchronize (compare results) at the end. The **synchronization** step is the comparison of the oldest pennies from the two bags, which ensures that the final result is the overall oldest penny.



Questions about this phase:

1. How did dividing the task between two people (parallel searching) affect the overall time compared to Phase 1 (sequential searching)? Did you experience speedup?
2. If each person worked at different speeds, how might that affect the overall time it takes to find the oldest coin?
3. Would adding more people (more sorters with smaller bags of coins) make this task faster? Why or why not?

Phase 3: Parallel Version B

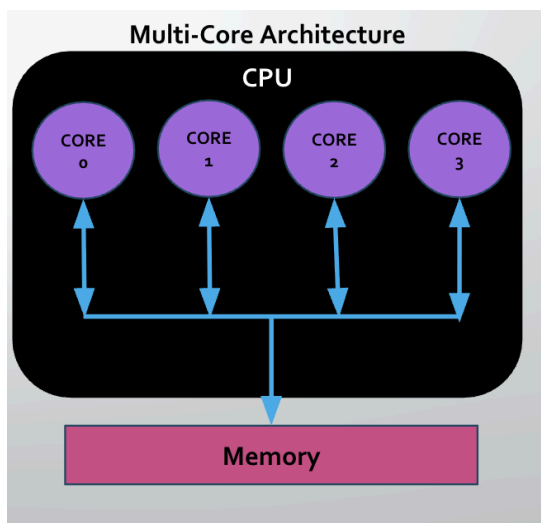
Select 1 person to be the timer (Person 1). All other people will be sorters. All sorter people except for 1 (Person 2), will be given four coins and all remaining coins will be given to Person 2.

All sorter people begin sorting for the oldest coin out of the coins provided for them. Then, the people must communicate with each other to produce the oldest coin out of all in the group.

The timer (Person 1) times and then records on the group lab sheet the total time it took to find the oldest coin.

What did PHASE 3 represent?

This phase introduces **data parallelism**, where the data (the coins) are divided among multiple workers. However, there is **load imbalance**, as one person has significantly more pennies to sort than the others. This imbalance reflects real-world scenarios where tasks are unevenly distributed among parallel workers, which can lead to inefficiencies. After sorting, the team members must communicate (**synchronize**) to combine their results and identify the oldest penny.



Phase 4: Parallel Version C

Imagine having enough people so that each person is given one coin. A person may only talk to one other person at a time (there is no yelling across the room). How long do you think it will take to find the oldest coin?

What did PHASE 4 represent?

This phase also represents **data parallelism**, where the data (coins) are divided among multiple workers (in fact, the data set is divided as much as it can be with each processor having only one coin). With the data being spread across so many processors, the synchronization step to merge the results ends up taking more time than the actual work, and this affects speed-up.

Turn in your answers to the following questions:

1. Continuing with the problem of searching a bag of coins for the oldest: If you had 100 pennies and could use up to 6 processors to search for the oldest coin, what strategy would be the most efficient? Assume that each processor is the same and that each processor can compare two coins at a rate of 1 per second. Assume that for the synchronization step, a comparison could occur at the rate of 1 per second. How long would the best strategy take?
2. Think about a task in real life that benefits from multiple people helping, but too many people will actually result in a slower performance.